Radio over Fiber (RoF)

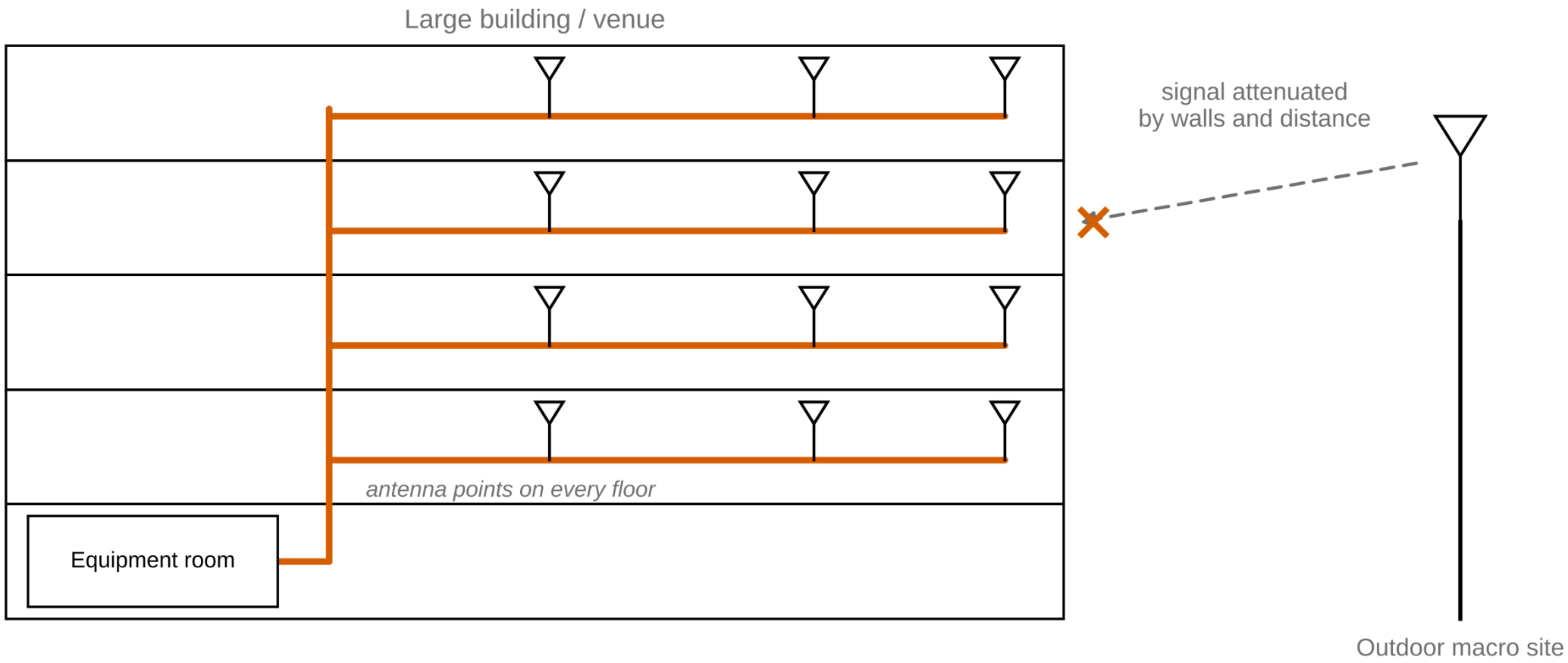
Carrying radio signals over optical fiber

Speaker A · Speaker B · 15 minutes



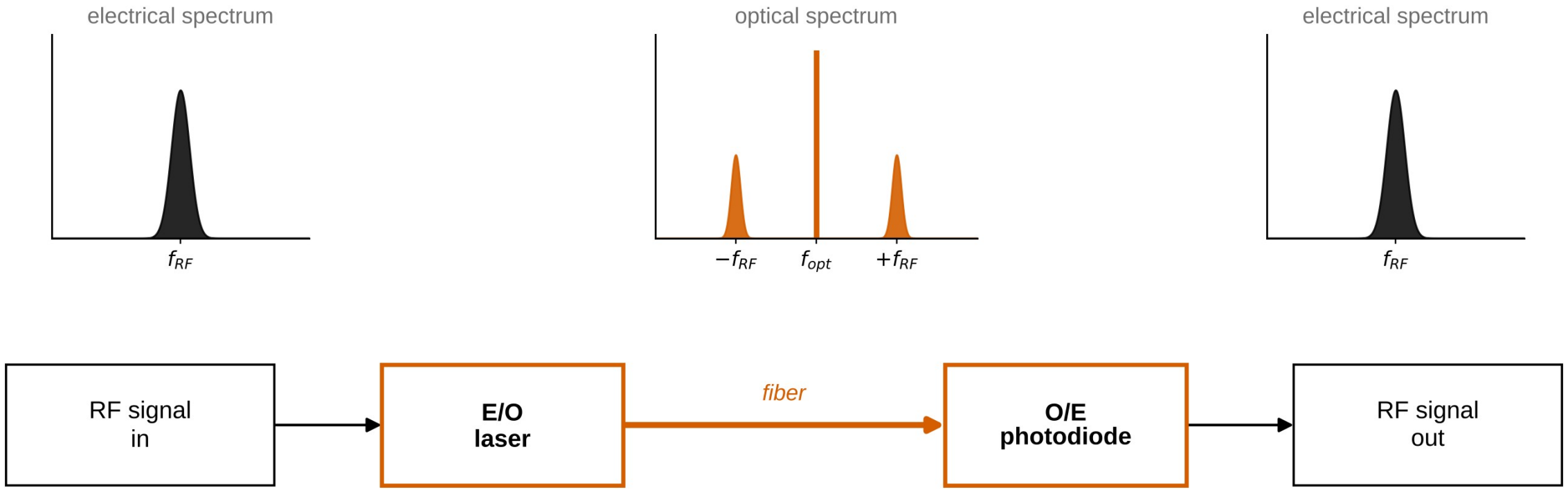
The problem

Large buildings and venues need many antenna points, fed from one equipment room.



What RoF is

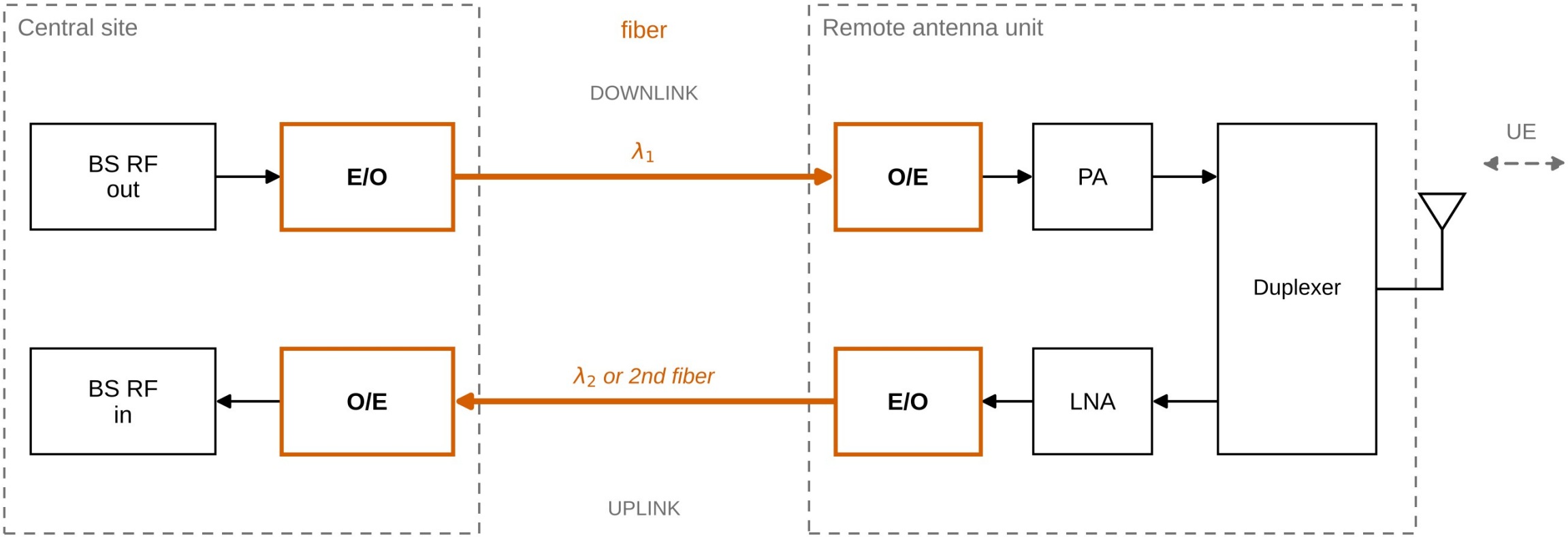
The radio waveform modulates light. A photodiode turns it back into the same radio signal.



Sources: ITU-T G.Suppl.55 (10/2025); ITU-T G.9803 (2018).

Signal path: downlink and uplink

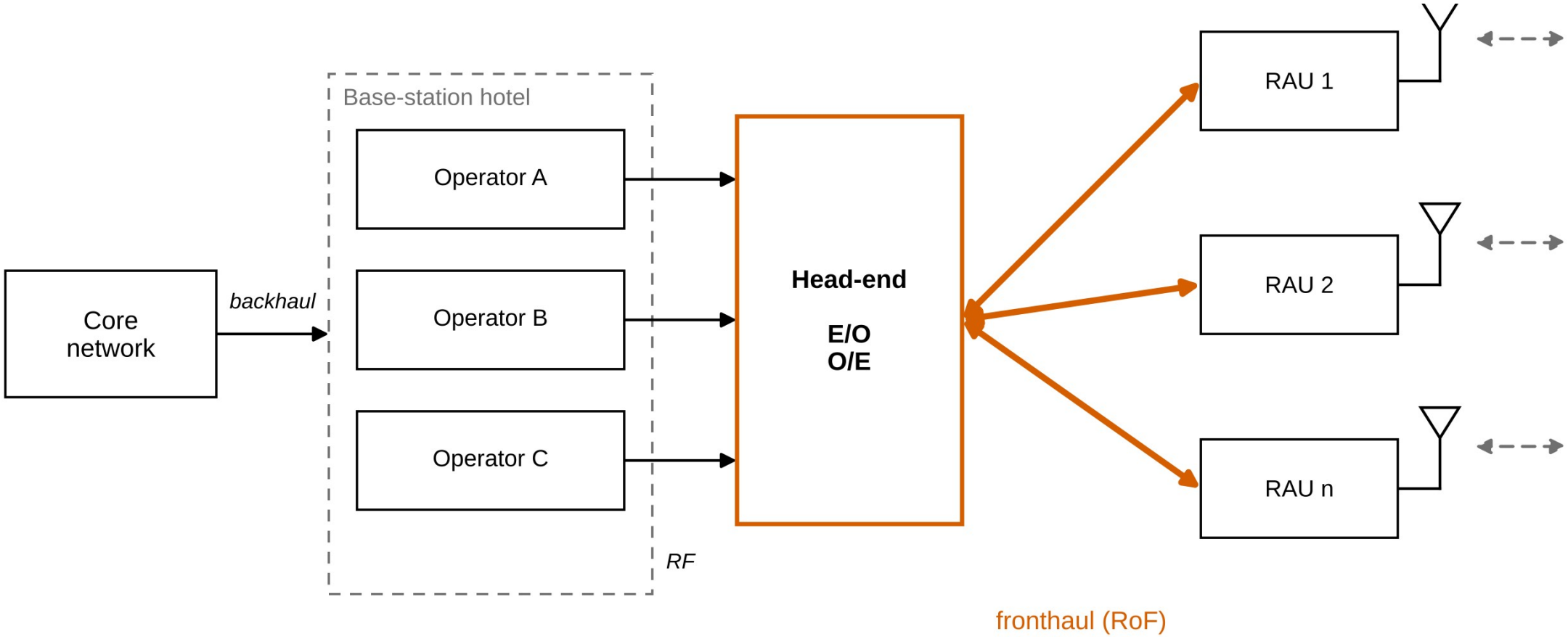
Four optical conversions in a two-way link. Each adds noise and distortion.



Source: C. H. Cox III, Analog Optical Links, Cambridge University Press, 2004.

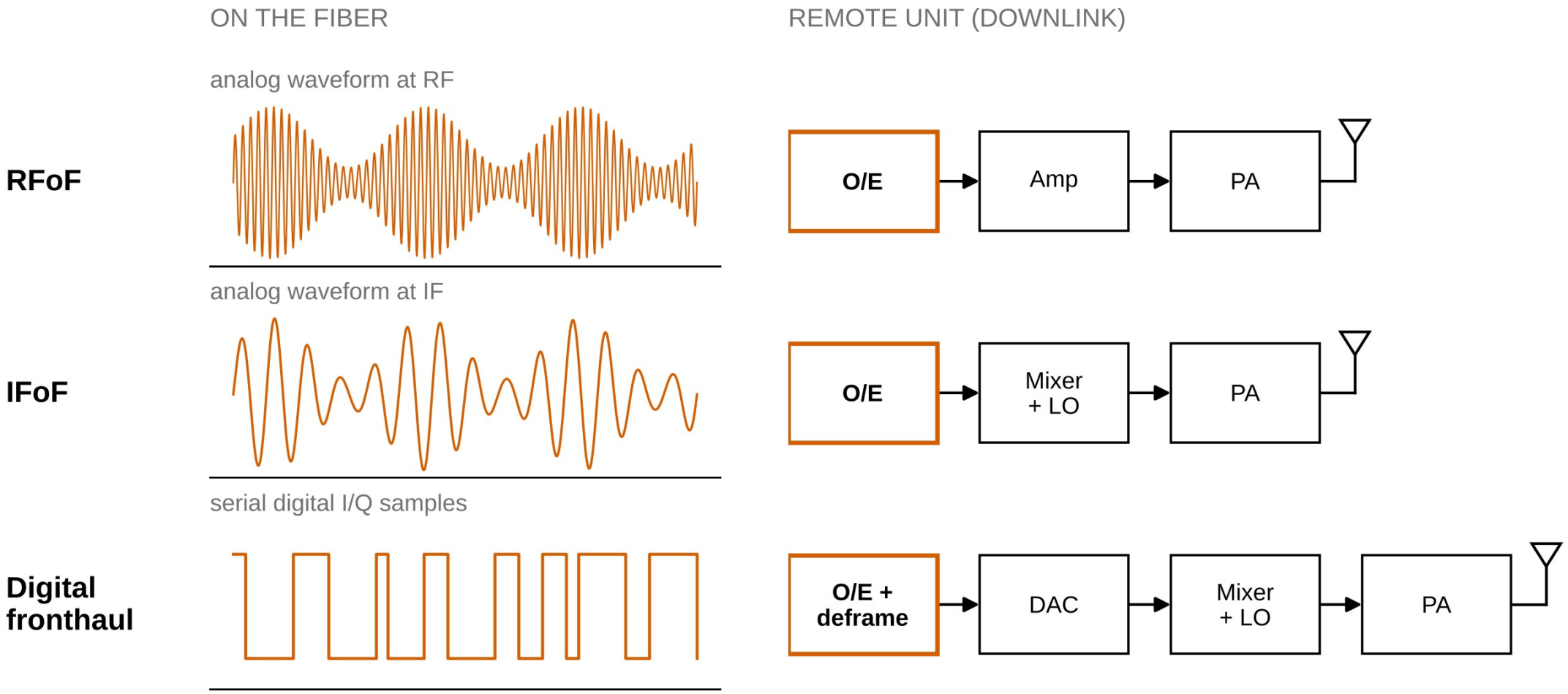
Where it sits in the network

RoF is the fronthaul link from head-end to remote unit. Backhaul stays digital.



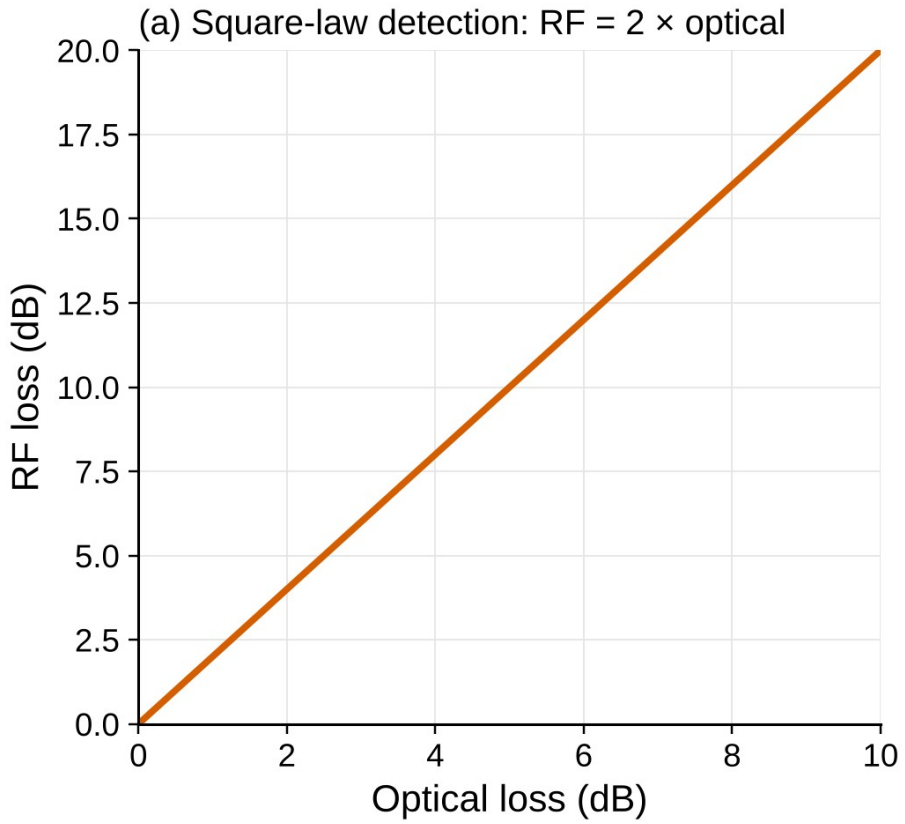
Three ways to use the fiber

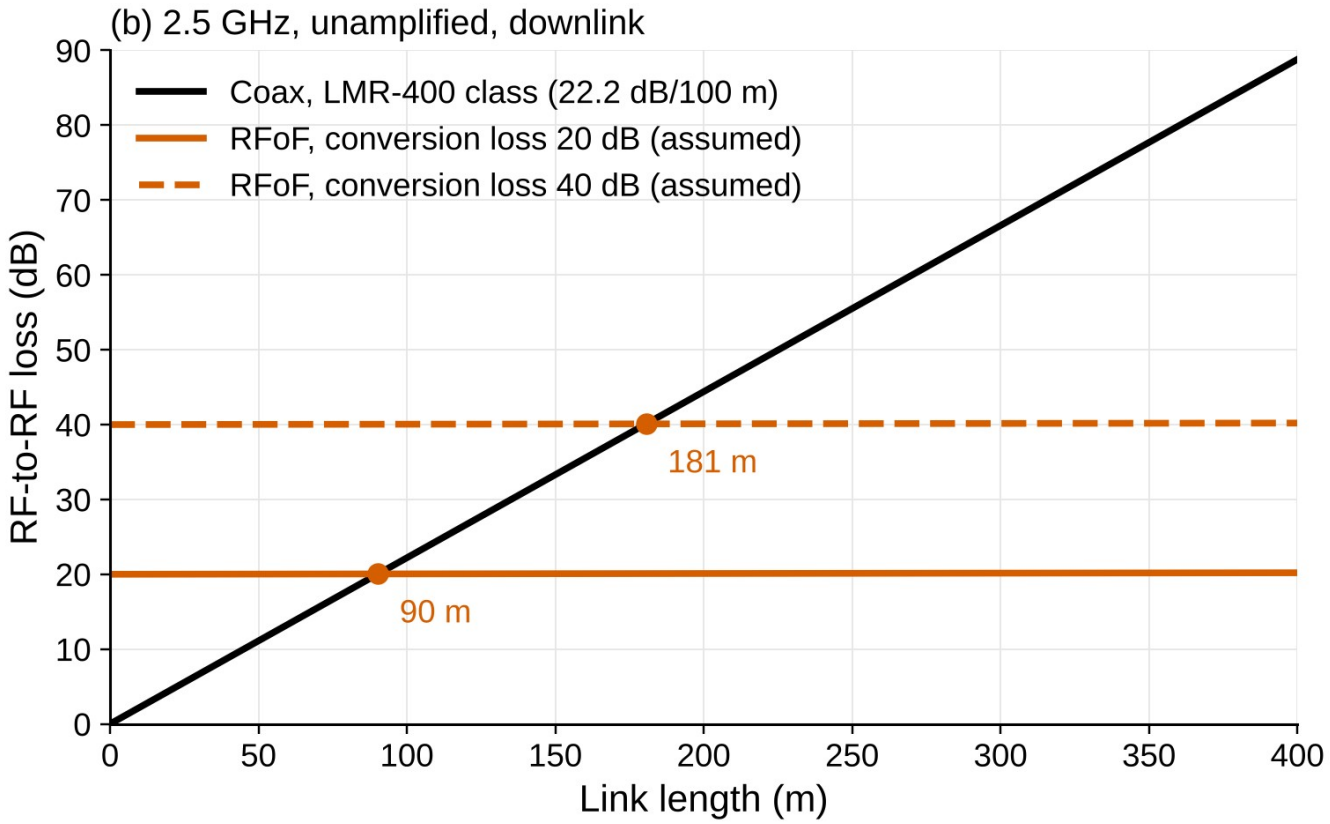
Simpler remote unit means analog signal on the fiber. Digital needs a full radio at the remote end.



Coax or fiber? It depends on length

Fiber has lower RF loss only beyond a break-even length set by the conversion loss.

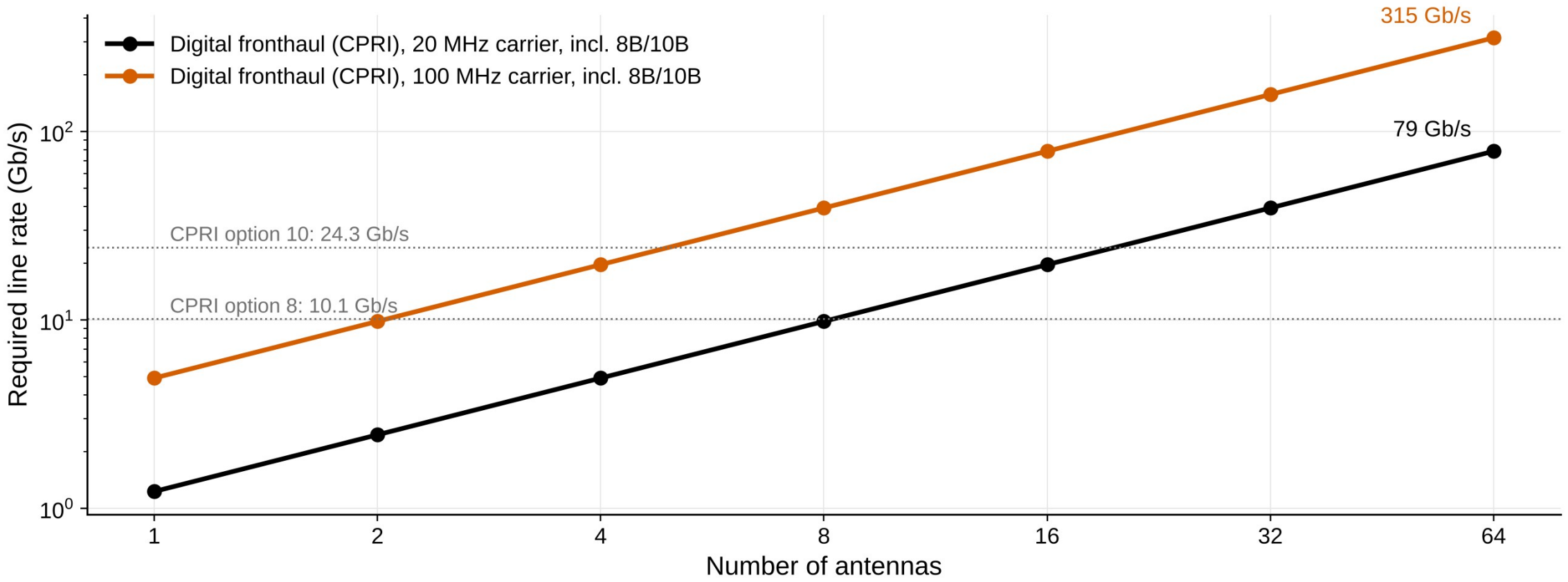




Coax: Times Microwave LMR-400 datasheet (22.2 dB/100 m at 2500 MHz). Fiber: G.652.D cable datasheets. Conversion loss values are assumptions.

Analog RoF vs digital fronthaul

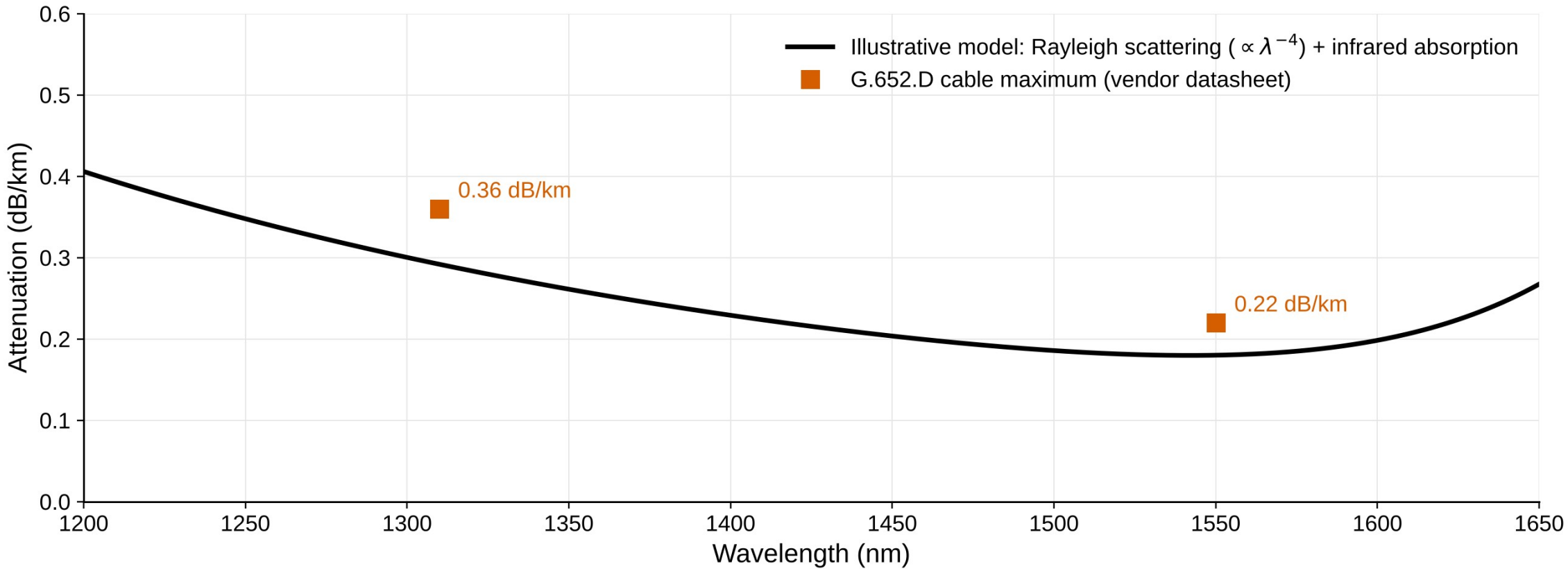
Digital fronthaul line rate grows with every antenna and every MHz of bandwidth.



Source: CPRI Specification V7.0 (2015). Rates computed with the formula in the notes.

Benefit: low fiber loss

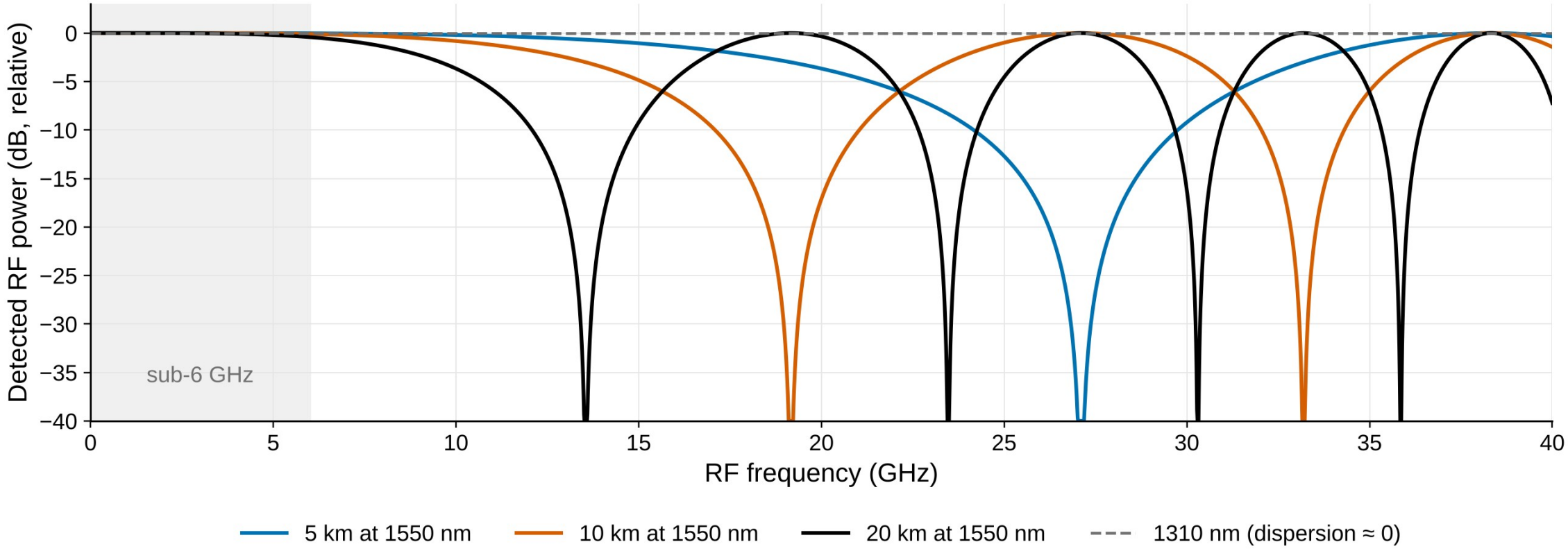
Fiber loss is a fraction of a dB per km, and the fiber itself does not pick up interference.



Datasheet points: Prysmian G.652.D cable. Model curve is illustrative (Rayleigh scattering and infrared absorption).

Limit 1: chromatic dispersion

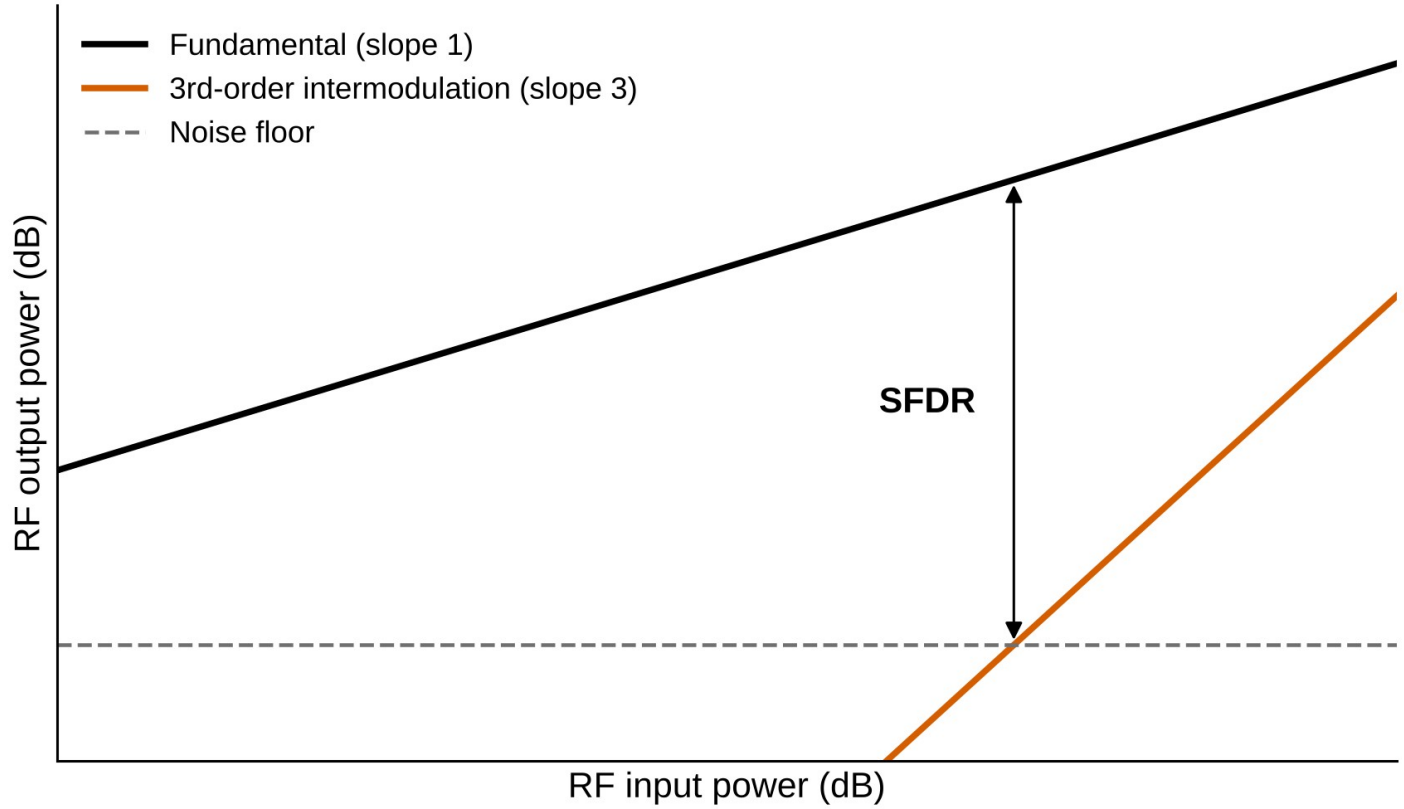
At 1550 nm the received RF power fades at certain frequencies. Sub-6 GHz bands are barely affected.



Model: U. Gliese et al., IEEE Trans. Microwave Theory Tech. 44(10), 1996. Curves computed for standard SMF.

Limit 2: noise vs distortion

An analog link has a limited usable power window.



Too little signal
lost in noise

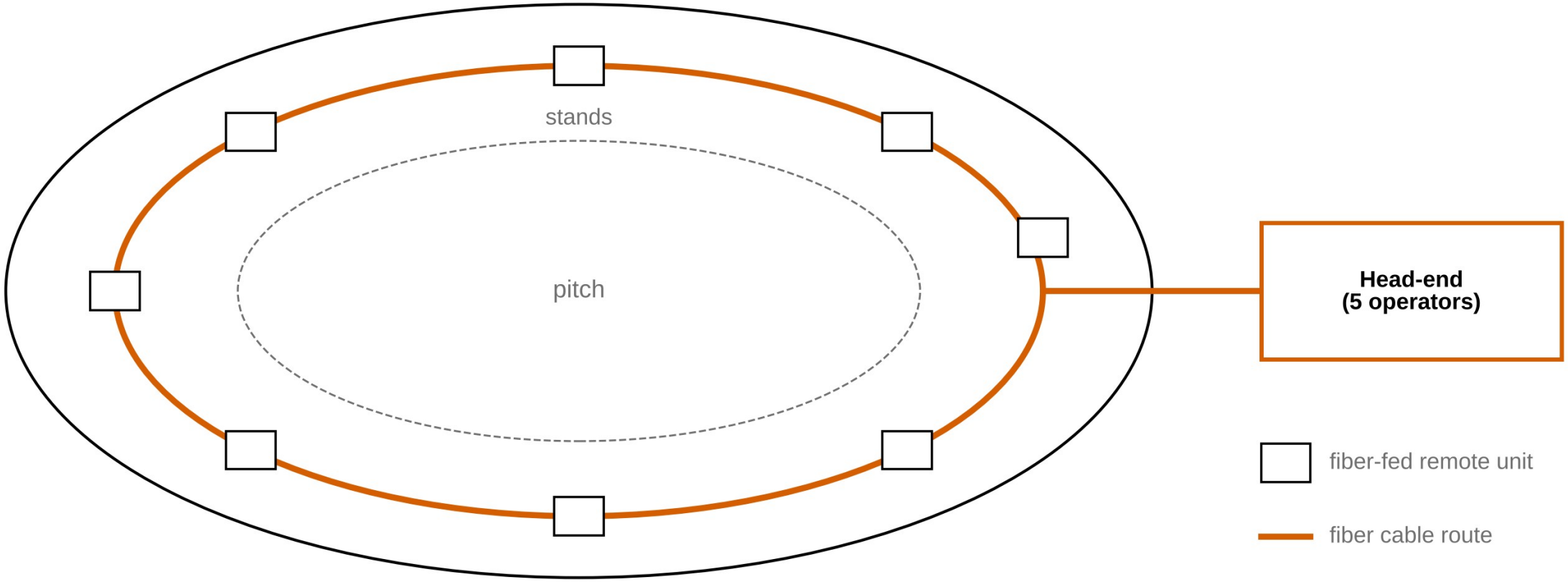
Too much signal
intermodulation appears

SFDR
the usable window between the two

Illustrative, no values. Source: C. H. Cox III, Analog Optical Links, 2004.

Case: Maracanã and Mineirão (2014)

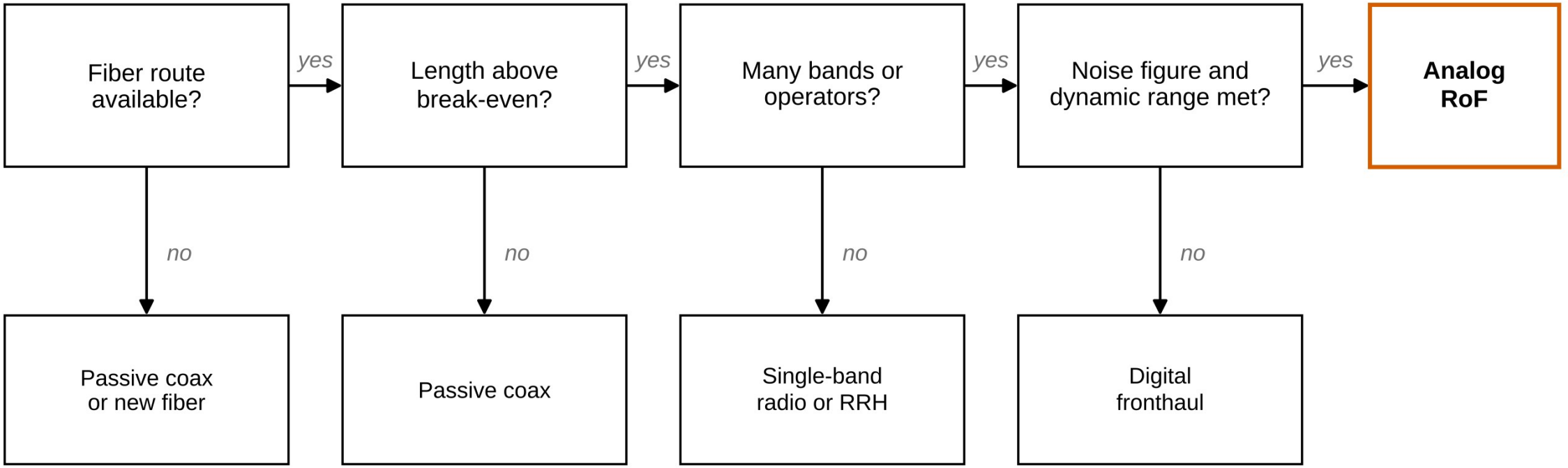
Fiber-fed distributed antenna system for five operators, 2G / 3G / 4G.



Sources: Axell Wireless press release, June 2014; RFS press release, 10 June 2014. Schematic only.

When to choose analog RoF

RoF moves the antenna point closer to users. It does not add spectrum.

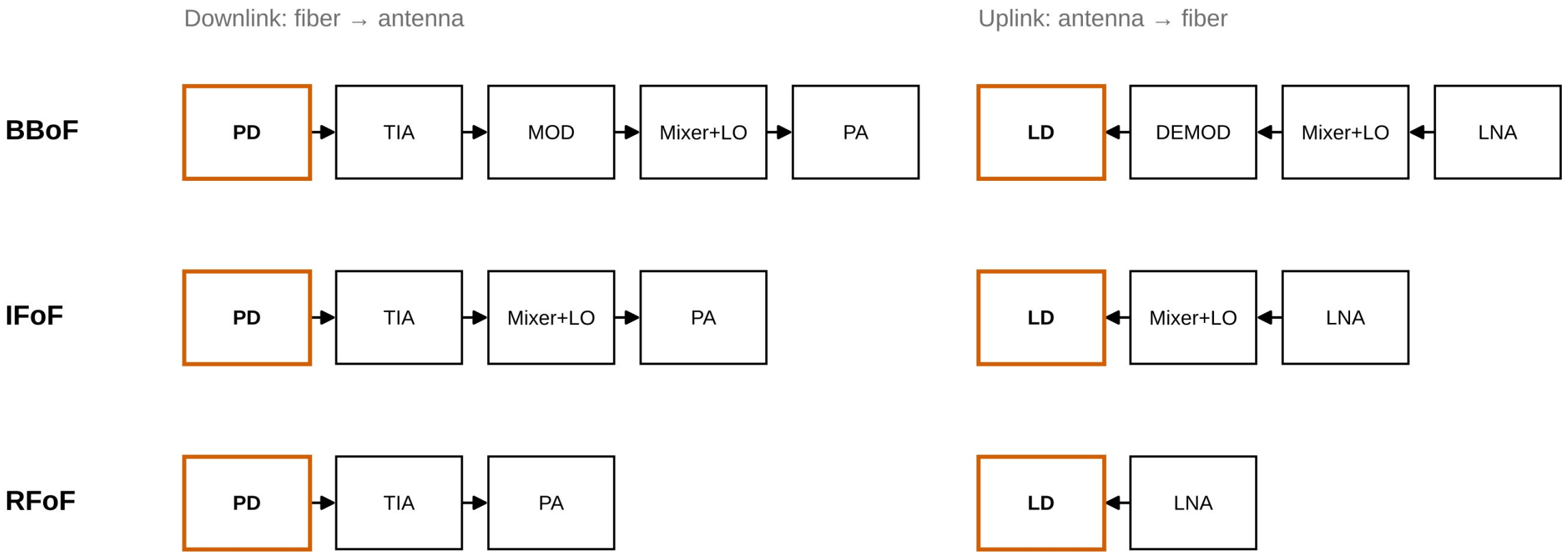


Questions



A1 Remote unit types

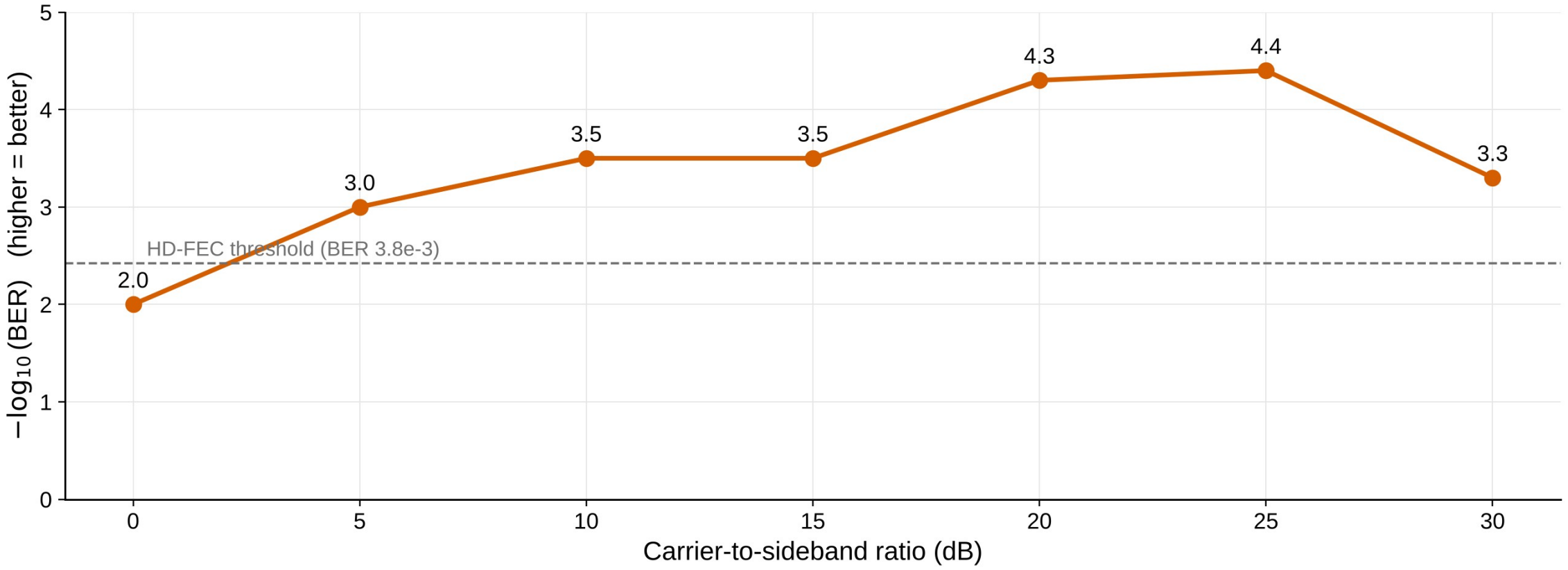
What the remote unit must contain for each signal on the fiber.



Redrawn from: Electronics 9(11), 1785 (2020), Fig. 3. DOI: 10.3390/electronics9111785.

A2 Measured BER vs carrier-to-sideband ratio

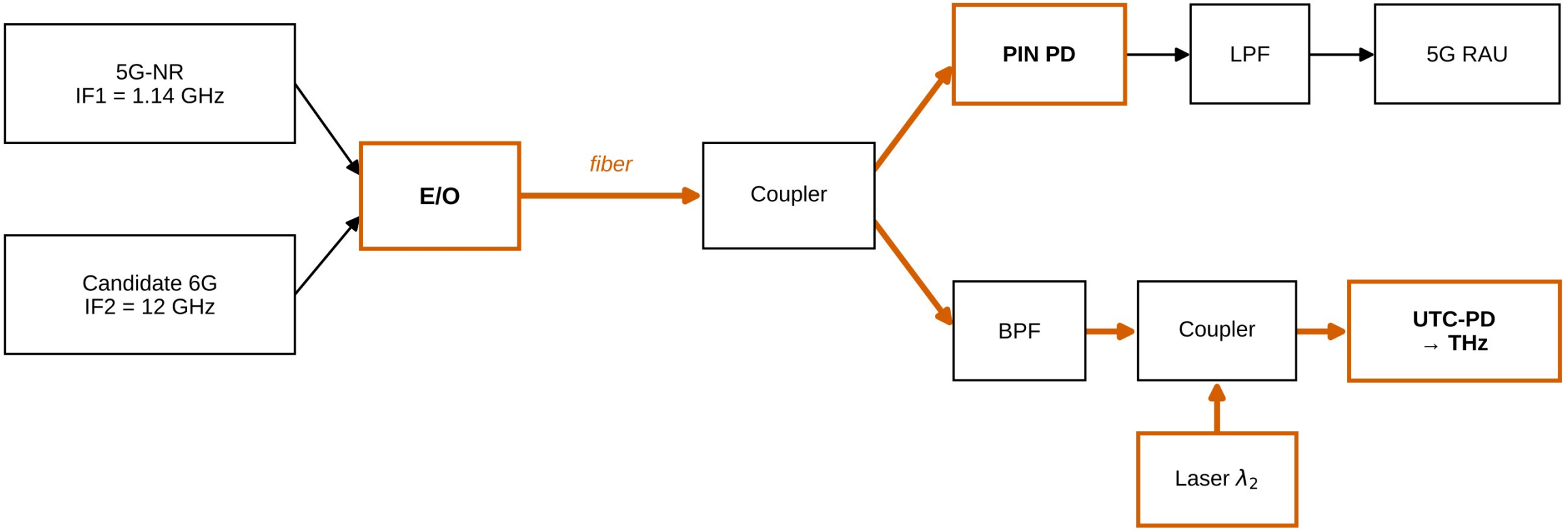
One experimental link: 10 km fiber + 3 m wireless. Best result near 25 dB.



Redrawn from: M. U. Hadi and G. Murtaza, Electronics 12(1), 64 (2023), Fig. 6. Values read from the published plot.

A3 Research demo: 5G + candidate 6G on one fiber

Laboratory demonstration, not a deployed system.



Redrawn and simplified from: M. U. Hadi and G. Murtaza, Electronics 12(1), 64 (2023), Fig. 2.

A4 References

1. C. H. Cox III, Analog Optical Links: Theory and Practice, Cambridge University Press, 2004.
2. U. Gliese, S. Nørskov, T. N. Nielsen, Chromatic dispersion in fiber-optic microwave and millimeter-wave links, IEEE T-MTT 44(10), 1996.
3. ITU-T G.9803 (2018), Radio over fibre systems, with Amendments 1 (2019) and 2 (2022).
4. ITU-T G-series Supplement 55 (10/2025), Radio-over-fibre technologies and their applications.
5. ITU-T G.652, single-mode optical fibre and cable; Prysmian G.652.D cable datasheet.
6. CPRI Specification V7.0 (2015); eCPRI Specification V2.0 (2019).
7. Y. Li, Q. Xie, M. El-Hajjar, L. Hanzo, Analogue Radio Over Fiber for Next-Generation RAN, arXiv:2111.13851, 2021.
8. Radio over Fiber: An Alternative Broadband Network Technology for IoT, Electronics 9(11), 1785, 2020.
9. M. U. Hadi, G. Murtaza, Fibre Wireless Distributed Antenna Systems for 5G and 6G Services, Electronics 12(1), 64, 2023.
10. Times Microwave Systems, LMR-400 datasheet.
11. Axell Wireless press release, June 2014; RFS press release, 10 June 2014.